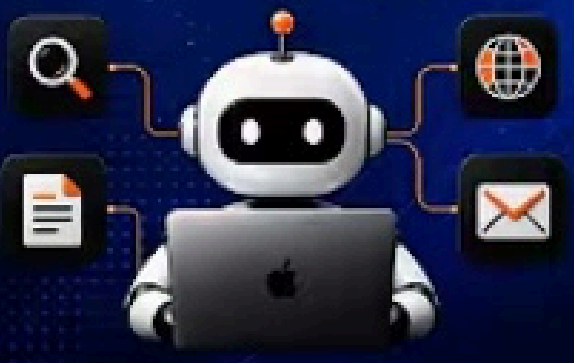
Complete Agentic AI Course 2026



- Fundamentals
- AI Agents & LLMs
- Core Agent Loop
- Tools & Function Callin
- Memory & Context
- RAG Explained
- Vector Databases
- Embeddings
- MCP Protocol
- Multi-Agent Systems
- Agentic Architectures
- Safety & Guardrails

Agentic AI Engineer Roadmap (2026)

Become an AIEngineer for the NextGenerationof Intelligent Systems

WhyAgenticAI?

Agentic AI is transforming software from static applications into autonomous systems that can:

- Reason
- Plan
- Use tools
- Collaborate
- Retrieve knowledge
- Execute workflows
- Learn from feedback

Modern AI Engineers are no longer just model builders. They are becoming architects of intelligent systems.

12-Stage Agentic AI Roadmap

Phase 1 — AI & Programming Foundations

Learn:

- Python fundamentals
- APIs & JSON
- Data structures
- Git & GitHub
- Linux basics
- REST APIs

Tools:

- Python
- VS Code
- GitHub
- Postman

Build:

- API-based chatbot
- Python automation scripts

Phase 2 — Machine Learning Basics

Learn:

- Supervised learning
- Unsupervised learning
- Regression & classification
- Evaluation metrics
- Feature engineering

Libraries:

- NumPy
- Pandas
- Scikit-learn
- Matplotlib

Build:

- Spam classifier
- Recommendation system
- Customer churn predictor

Phase 3 — Deep Learning & Neural Networks

Learn:

- Neural networks
- Backpropagation
- CNNs
- RNNs
- Transformers

Frameworks:

- PyTorch

- TensorFlow

Build:

- Image classifier
- Text classifier
- Sentiment analysis model

Phase 4 — LLM Fundamentals

Learn:

- Transformers
- Attention mechanism
- Embeddings
- Tokenization
- Prompt engineering
- Context windows
- Hallucinations

Platforms:

- OpenAI
- Claude
- Gemini
- Hugging Face

Build:

- AI assistant
- Prompt workflows
- Summarization system

Phase 5 — RAG (Retrieval-Augmented Generation)

Learn:

- Vector databases
- Semantic search
- Chunking
- Embeddings

- Re-ranking
- Hybrid retrieval

Tools:

- Pinecone
- ChromaDB
- Weaviate
- FAISS

Build:

- PDF chatbot
- Knowledge base assistant
- Internal documentation bot

Phase 6 — AI Agents & Agentic Systems

Learn:

- Single-agent systems
- Multi-agent systems
- Planning & execution
- Tool calling
- Memory systems
- Reflection loops
- Autonomous workflows

Frameworks:

- LangChain
- LangGraph
- CrewAI
- AutoGen
- OpenAI Agents SDK

Build:

- Research agent
- Multi-agent workflow
- Autonomous task planner

Phase 7 — MCP (Model Context Protocol)

Learn:

- Context sharing
- External tool integration
- Secure agent communication
- Structured context systems

Build:

- MCP-enabled AI workflow
- AI workspace assistant

Phase 8 — AI Memory Systems

Learn:

- Short-term memory
- Long-term memory
- Vector memory
- User personalization
- Session persistence

Build:

- Personalized AI assistant
- Memory-enabled chatbot

Phase 9 — AI System Design

Learn:

- AI architecture
- Scalability
- Latency optimization
- Cost optimization
- AI orchestration
- Event-driven systems

Topics:

- Agent orchestration
- AI pipelines
- Queue systems
- Caching
- Observability

Build:

- Production-grade AI workflow
- AI orchestration dashboard

Phase 10 — AI Deployment & MLOps

Learn:

- FastAPI
- Docker
- Kubernetes basics
- CI/CD
- Monitoring
- Clouddeployment

Platforms:

- AWS
- Azure
- GCP
- Hugging Face Spaces

Build:

- AI API service
- Deploy RAG application
- Cloud AI agent

Phase 11 — AI Governance & Safety

Learn:

- AI guardrails
- Prompt security
- AI governance
- Human-in-the-loop systems
- Monitoring & auditing

Build:

- Safe AI workflow
- Moderation system
- AI validation pipeline

Phase 12 — Real-World AI Projects

Duration: Ongoing

Build Portfolio Projects:

Beginner:

- Resume analyzer
- AI note summarizer
- PDFchatbot

Intermediate:

- Multi-agent research assistant
- AI automation dashboard
- AI email assistant

Advanced:

- Enterprise RAG system
- Autonomous AI operations agent
- Multi-agent orchestration platform
- AI coding assistant

Recommended Tech Stack

Programming

- Python
- SQL
- JavaScriptbasics

AI Frameworks

- LangChain
- CrewAI
- AutoGen
- LangGraph

- HuggingFace

Databases

- PostgreSQL
- Pinecone
- ChromaDB
- Redis

Deployment

- FastAPI
- Docker
- AWS
- Vercel

Monitoring

- LangSmith
- Weights & Biases
- Grafana

Free Learning Resources

Courses

- GoogleAI Courses
- Microsoft AI for Beginners
- DeepLearning.AI

- Hugging Face Course
- OpenAI Cookbook

Practice Platforms

- Kaggle
- GitHub
- Hugging Face Spaces

Career Roles After This Roadmap

You can apply for:

- AI Engineer
- Generative AI Engineer
- LLM Engineer
- AI Automation Engineer
- Agentic AI Engineer
- AI Product Engineer
- AI Solutions Architect
- Applied AI Engineer

Final Advice

Do not spend all your time consuming tutorials.

The fastest way to become an AI Engineer is:

Learn → Build → Deploy → Iterate

Projects matter more than certificates.

The future belongs to engineers who can combine:

- AI
- Workflows
- Agents
- Automation
- System Design
- Human + AI collaboration

Agentic AIEngineer Roadmap

Learn the future of AI systems engineering.

Build intelligent systems that can reason, retrieve, plan, collaborate, and execute autonomously.

Follow @AkhilMeesala for more AI Engineering and Agentic AI content.